

Distributed Dispatch of Integrated Electricity-Heat Systems With Variable Mass Flow

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Abstract—The constant mass flow assumption has dominated distributed dispatch of integrated electricity-heat systems (IEHSs), which ensures the simplicity of decomposition while incurring opportunity costs. In contrast, a heat operation strategy with variable flow and variable temperature (VF-VT) enhances flexibility and optimality. However, VF-VT renders the IEHS dispatch problem into a mixed-integer nonlinear bi-level nested program, which leaves a critical yet unresolved challenge for distributed autonomous dispatch. Therefore, this paper proposes a two-stage alternating procedure embedded with sequential equivalent techniques. A feasible initial point is obtained in the first stage, and the total costs are minimized thereafter. In each iteration, the heat sector optimizes both hydraulic and thermal states based on a surrogate model, and submits the heat equivalent to the electricity sector; the electricity sector solves the reduced IEHS dispatch problem and then updates the boundary. The feasibility is proved theoretically, while numerical tests validate the effectiveness.

Index Terms—Integrated energy system, combined heat and power generation, variable mass flow, mixed-integer nonconvex programming, distributed optimization.

NOMENCLATURE

Indices

b, t Index of pipelines in district heating networks (DHNs) and time slots, respectively.

Parameters

$\underline{\tau}/\bar{\tau}$ Upper/lower bounds of thermal limit.
 δ_t Hydraulic limit at period t .
 F_E, B_E Coefficient matrices.
 Q_E, A Right-hand side coefficient vector.
 d_E, d_{TH}, a_H Coefficient of pressure loss in pipeline b .
 μ_b Density of water.
 ρ Specific heat capacity of water.

$L_b/A_b/\lambda_b$ Length/cross-sectional area/heat transfer coefficient of pipeline b .
 $n1/n2$ Starting and ending nodes of pipeline b .
 L Number of iterations at the end of the first stage.
 $T_{h,t}^{out}$ Outdoor temperature of house h at period t
 α_h, β_h Thermal parameters of the environment and the distributed heat pump of house h .

Variables

x_E, x_H Internal decision variables of electric power networks and DHNs, respectively.
 y_E, y_H Boundary decision variables of EPNs and DHNs, respectively.
 p^{EHC} Electricity output of electricity-heat coupling units.
 h^{EHC} Heat output of electricity-heat coupling units.
 d^{PUMP} Electric power consumption of water pump.
 ps_1 / pr_1 Pressure head at the supply/return end of the reference node.
 ps_{-1} / pr_{-1} Pressure head at the supply/return end of nodes other than the reference node
 m Mass flow rate.
 τ Mass flow temperature.
 ξ Complicating dynamic variables addressing transfer delays and heat losses in DHNs.
 $\kappa/v, \omega$ Slack variables for the lower/upper thermal limit and the hydraulic limit, respectively.
 $\gamma_{b,t}, \varphi_{b,t}$ Numbers of periods denoting time delays of pipeline b at period t associating changes in temperature.
 $R_{b,t}, S_{b,t}$ Coefficient variables of pipeline b at period t associated with the historic flow mass.
 $J_{b,t}$ Heat loss coefficient of pipeline b at period t .
 $K_{b,t,k}$ The k -th coefficient variables defining the outlet temperature of pipeline b at period t .
 $d_{h,t}$ Power of distributed heat pump in house h at period t
 $T_{h,t}^{in}$ Indoor temperature of house h at period t

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I. INTRODUCTION

THE FORTHCOMING crisis arising from the depletion of primary energy urges the need for energy transition, while the integrated energy system has been recognized as a cornerstone in industrial systems to pave the way toward carbon neutrality. In the integrated electricity-heat system (IEHS),

a typical form of the integrated energy system, the flexible operation of district heating networks (DHNs) contributes to more accommodation of renewables in electric power networks [1]. In Jilin province in Northeast China, a pilot project on IEHS dispatch initiated by the local electric power network operator has helped accommodate 567 million kWh of wind power over the recent three years. Such contribution is expected to keep rising considering the annual growth rate of the DHN scale, e.g., 10-15% in China since 1998 [2].

As for architecture, centralized dispatch is not compatible with the distributed structure of IEHSs, while distributed dispatch is more realistic to achieve synergy among different energy sectors. Generally, electric power networks and DHNs are invested, owned and operated by different entities, where operational independence and individual privacy must be taken care of [3]. In reality, it will be impractical for each entity to disclose its data and models to others as required by centralized IEHS dispatch. Therefore, remarkable efforts have been made lately to distributed IEHS dispatch by employing, e.g., decomposition-coordination heuristic solution [4], Lagrangian relaxation [5], subgradient method [6], alternating direction of multipliers [7], optimality condition decomposition [8], Benders decomposition [9], [10], [11], equivalent-based approach [12], [13] and neurodynamic-based approach [14].

Nonetheless, mass flow in DHNs is assumed to be constant in most distributed IEHS dispatch research [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], which simplifies the decomposition procedure while compromising economic benefits. Constant mass flow refers to fixing the mass flow rate, while adjusting the mass flow temperature to meet the heat load demand. This is also known as the constant flow and variable temperature (CF-VT) operation. The story behind is that the DHN constraints are linear under CF-VT. Then the resultant IEHS dispatch problem is convex, which is a critical prerequisite for the application of most distributed optimization algorithms [16]. But fixing the mass flow drastically undermines the flexibility of DHNs; technically, the constant mass flow assumption shrinks the feasible region of the IEHS dispatch problem, thus impairing the economic optimality and robustness of the system. Although demand response may be initiated, the quality of heat supply is compromised and the required incentives incur additional costs.

To unleash the flexibility of DHNs, a heat operation strategy with variable flow and variable temperature (VF-VT) is more favorable. Nonetheless, VF-VT complicates the decomposition solution procedure significantly. Considering the true dynamics of DHNs, VF-VT renders the IEHS dispatch problem into a bi-level nested mixed-integer nonlinear program (BN-MINLP) [17]. Based on fixed-point iteration, a centralized heuristic solution is introduced for IEHS dispatch under VF-VT in [18], but the DHN constraints are relaxed during the procedure and thus the final dispatch decision is not feasible. This complex problem is technically demanding even in centralized dispatch where global information is available and attracting endeavors to date such as tightening McCormick method [19] or convex relaxation techniques [20], let alone in

distributed dispatch where only local information and boundary states are available in regional subproblems due to privacy concerns. Scant early attempts can be found in the distributed dispatch of IEHSs under VF-VT, which generally neglect or oversimplify dynamics in DHNs. A coarse-grained IEHS dispatch model is decomposed in [4], where heat loss is estimated by experience. A best-response decentralized algorithm is devised in [21] using a market-based approach. These works have neglected heat transfer delay, the main physical feature of DHNs, to approximate the original BN-MINLP as a single-level continuous problem so that commercial solvers are directly applicable. Besides, they simply assume that, given any boundary condition or price signal, regional DHN subproblems are always feasible. In fact, it remains a significant challenge to obtain a feasible solution in distributed IEHS dispatch under VF-VT, as shown by the concrete examples in simulation results that refute the above assumption.

There is some decentralized IEHS market operation research addressing heterogeneous interests of EPNs and DHNs by using a non-cooperative game; see, e.g., [22] and our previous attempt [23]. However, the non-cooperative game might lead to economic efficiency losses. Hence, this paper follows the cooperative paradigm of distributed dispatch as [4], [5], [6], [7], [8], [9], [10], [11], [13], [14], [16], and proposes a distributed solution for the IEHS dispatch problem with the variable mass flow, which is our main contribution. To our best knowledge, this is the first distributed solution for the IEHS dispatch problem under VF-VT with full consideration of heat quasi-dynamics, which is essentially an intractable BN-MINLP. Specific contributions of this paper include:

(1) A two-stage alternating procedure is established for distributed dispatch, where the first stage finds a feasible initial point of the original BN-MINLP, while the second stage minimizes the total costs in the IEHS. Each stage involves solving iteratively an electric power network subproblem and a DHN subproblem. The feasibility of this procedure is proved mathematically. The advantages over the centralized fixed-point iterative solution in [18] are discussed in depth both theoretically and numerically.

(2) In the electric power network subproblem, a sequential equivalent method for DHNs under VF-VT is proposed, which projects the feasible region defined by both thermal and hydraulic constraints onto the boundary. Information privacy of DHNs is preserved, while computation efficiency is enhanced by reducing the total number of variables and constraints.

(3) In the DHN subproblem, a surrogate model is investigated to approximate heat dynamics including transfer delays and heat losses in DHNs. The optimization of mass flow is facilitated by resorting to a single-level continuous nonlinear program rather than the original BN-MINLP.

The remainder of the paper is organized as follows. IEHS dispatch under VF-VT is analyzed in Section II. The two-stage alternating procedure is devised in Section III. Section IV formulates the electric power network subproblem in conjunction with the sequential equivalent method for DHNs under VF-VT. The DHN subproblem is simplified by the surrogate

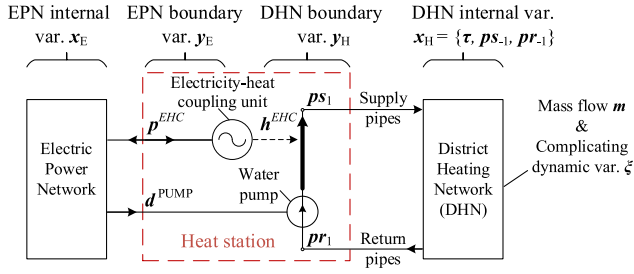


Fig. 1. Distributed structure of an IEHS under VF-VT.

dynamic model in Section V. Case studies are carried out in Section VI, while Section VII concludes the paper.

II. INTEGRATED ELECTRICITY-HEAT DISPATCH WITH VARIABLE FLOW

The model of IEHS dispatch under VF-VT is introduced first, and then the difficulties in decomposition are identified by analyzing the features of the model.

A. Mathematical Model

For sake of completeness, the elaborate form of the centralized IEHS dispatch model under VF-VT is introduced in Appendix A. It is reformulated in the context of distributed dispatch here, while the connection between the compact and the elaborate versions is pointed out so that interested readers may easily get full details of the model by referring to Appendix A.

Before the reformulation, variables are defined and partitioned as shown in Fig. 1. y_E is the electric power network boundary decision vector including electricity output of electricity-heat coupling units p^{EHC} and electric power consumption of water pump d^{PUMP} . Note that electricity-heat coupling units refer to the devices at the electricity-heat coupling boundary such as combined heat and power (CHP) units and heat pumps. For a heat pump, its electricity output is negative, which means electricity consumption. x_E represents the electric power network internal decision vector including electricity output and upward/downward spinning reserve capacity of non-CHP units, phase angle of buses, electricity output of wind farms. $ps_{n,t}/pr_{n,t}$ denotes the pressure head at node n , period t in the supply/return pipes. The DHN boundary decision vector y_H encompasses heat output h^{EHC} and pressure head at the supply/return end of water pump ps_1/pr_1 which collects $ps_{1,t}/pr_{1,t}$ from all time slots at the reference node. Generally, a node with a heat station at the electricity-heat coupling boundary is selected to facilitate the calculation. A heat station consists of heat sources (CHP units or large-scale heat pumps) and pressure sources (water pumps). Mass flow temperature τ and pressure head at the remaining nodes ps_{-1}/pr_{-1} in the DHN are enclosed in the DHN internal decision vector x_H . $m_{b,t}$ denotes the mass flow rate of pipeline b at period t , which is concatenated into a decision vector in the following way $m \triangleq \text{col}(m_t) \triangleq \text{col}[\text{col}(m_{b,t})]$ where $\text{col}(\cdot)$ is the concatenation operator. ξ represents the complicating dynamic vector dependent on m . Both m and ξ are excluded

from x_H since they are special variables that contribute complexity to the model. Q_H , F_H , B_H and d_{HS} are coefficients to be determined by m and ξ , which are all variables under VF-VT.

A ready-to-decompose model (P0) is introduced and summarized:

$$(P0) \quad \min_{x_E, x_H, y_E, y_H, m, \xi, Q_H, d_{HS}, F_H, B_H} C(x_E, y_E, y_H) \quad (1)$$

$$\text{s.t. } \xi = \Phi(m) \quad (2)$$

$$F_E x_E + B_E y_E \leq d_E \quad (3)$$

$$Q_E \cdot y_E + Q_H(m) \cdot y_H \leq d_{HS}(m) \quad (4)$$

$$F_H(m, \xi) \cdot \tau + B_H(m, \xi) \cdot h^{EHC} = d_{TH} \quad (5)$$

$$\underline{\tau} \leq \tau \leq \bar{\tau} \quad (6)$$

$$\begin{cases} ps_{n1,t} - ps_{n2,t} = \mu_b(m_{b,t})^2 \\ pr_{n1,t} - pr_{n2,t} = \mu_b(m_{b,t})^2 \end{cases} \quad \forall t, \quad \forall b : n1 \rightarrow n2 \quad (7)$$

$$Am \leq a_H \quad (8)$$

$$ps_{-1,t} - pr_{-1,t} \geq \underline{\delta}_t, \quad \forall t \quad (9)$$

where the *non-elementary* function $\Phi(\cdot)$ is elaborated as follows:

$$\gamma_{b,t} = \arg \min_{n \in \mathbb{N}} \left\{ n : \text{s.t. } \sum_{k=0}^n (m_{b,t-k} \cdot \Delta t) \geq \rho A_b L_b \right\} \quad (10)$$

$$\phi_{b,t} = \arg \min_{n \in \mathbb{N}} \left\{ n : \text{s.t. } \sum_{k=1}^n (m_{b,t-k} \cdot \Delta t) \geq \rho A_b L_b \right\} \quad (11)$$

$$R_{b,t} = \sum_{k=t-\gamma_{b,t}}^t (m_{b,k} \cdot \Delta t) \quad (12)$$

$$S_{b,t} = \begin{cases} \sum_{k=t-\phi_{b,t}+1}^t (m_{b,k} \cdot \Delta t), & \text{if } \phi_{b,t} \geq \gamma_{b,t} + 1 \\ R_{b,t}, & \text{otherwise} \end{cases} \quad (13)$$

$$J_{b,t} = \exp \left[-\frac{\lambda_b \Delta t}{A_b \rho c} \left(\gamma_{b,t} + \frac{1}{2} + \frac{S_{b,t} - R_{b,t}}{m_{b,t-\gamma_{b,t}} \cdot \Delta t} \right) \right] \quad (14)$$

$$K_{b,t,k} = \begin{cases} (m_{b,t} \cdot \Delta t - S_{b,t} + \rho A_b L_b) / (m_{b,t} \cdot \Delta t), & k = t - \phi_{b,t} \\ (m_{b,k} \cdot \Delta t) / (m_{b,t} \cdot \Delta t), & k = t - \phi_{b,t} + 1, \dots, t - \gamma_{b,t} - 1 \\ (R_{b,t} - \rho A_b L_b) / (m_{b,t} \cdot \Delta t), & k = t - \gamma_{b,t} \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

The complicating constraint (2) originates from the node method [17], a canonical way to accurately describe temperature quasi-dynamics of pipelines, which has been widely used in DHN analysis; just list a few, [4], [7], [10], [12], [13], [24]. It specifies how heat dynamics are determined by the mass flow. As shown by (10)/(11), $\gamma_{b,t}/\phi_{b,t}$ indicates the past time slots where water mass outflows pipeline b before the end of period $t/(t-1)$. Defined by (12)/(13), $R_{b,t}/S_{b,t}$ represents the total mass flowing into pipeline b from period $(t-\gamma_{b,t})/(t-\phi_{b,t})$ to t . $J_{b,t}$ is the heat loss factor variable obtained in (14). Calculated by (15), $K_{b,t,k}$ is the heat transfer delay variable of pipeline b that maps the inlet temperature at period k onto the outlet temperature at period t . The above heat dynamic variables are collected by the complicating decision vector $\xi = \{K_{b,t,k}, J_{b,t}, \gamma_{b,t}, \phi_{b,t}, R_{b,t}, S_{b,t} \mid \forall b, t, k\}$.

Parameters are explained as follows. ρ is the density of water, while c refers to the specific heat capacity of water. L_b , A_b and λ_b are the lengths, the cross-sectional area, and the heat

transfer coefficient of pipeline b , respectively. Symbols with upper/lower bars, e.g., $\underline{\tau}$, $\bar{\tau}$, $\underline{\delta}_t$, denote upper/lower bounds of thermal and hydraulic conditions, respectively. μ_b is the pressure loss coefficient of pipeline b . $n1$ and $n2$ denote the starting and ending nodes of pipeline b , respectively. $\mathbf{d}_{TH}$ is the right-hand side coefficient vector of the thermal equations (mainly determined by heat load). $\mathbf{F}_E$, $\mathbf{B}_E$, $\mathbf{d}_E$, $\mathbf{Q}_E$, $\mathbf{A}$ and $\mathbf{a}_H$ are coefficient matrices/vectors that can be readily determined by referring to the mapping introduced below.

Objective (1) minimizes the total cost comprising of the operation cost of CHP/non-CHP units and the penalty cost of wind power spillage, which is convex and detailed by (40) in Appendix A. Constraint (3) refers to the electric power network constraints detailed by (44)-(50). Constraint (4) refers to the heat station constraints detailed by (51)-(55). Constraint (5) refers to the thermal constraints in the DHN detailed by (57) and (66)-(73). Constraint (6) refers to the temperature limits in the DHN detailed by (56) and (58). Constraint (7) is the Darcy-Weisbach equation rephrased from (64)-(65). Constraint (8) includes the mass flow continuity constraint and the mass flow rate limit as specified by (60)-(61) and (62)-(63), respectively. Constraint (9) refers to the pressure operational constraint detailed by (59). The direction of mass flow is known in our paper, which follows the convention established by the existing VFVT research, say, [18], [19], [20]. For a more detailed explanation of the centralized IEHS dispatch model under VF-VT, interested readers can refer to Appendix A.

B. Complexity Analysis: Challenges in Decomposition

Under the CF-VT operation strategy, ξ , $\mathbf{Q}_H$, $\mathbf{F}_H$, $\mathbf{B}_H$ and $\mathbf{d}_{HS}$ all become constants determined by $\mathbf{m}$. The DHN constraint is linear, resulting in a convex IEHS dispatch model, which is the prerequisite of most distributed optimization algorithms. Subsequently, mature distributed optimization methods such as the alternating direction of multipliers [7] and Benders decomposition [9], [10], [11] can be readily adopted. However, decomposition for IEHS dispatch under VF-VT is another story.

First, the VF-VT model (P0) is intractable even in a centralized manner, which cannot be directly solved by off-the-shelf solvers such as Gurobi and IPOPT. Apart from the discreteness introduced by $\gamma_{b,t}$ and $\varphi_{b,t}$ and the nonconvexity arising from constraints (4), (5) and (7), the model (P0) is much more complicated than a traditional mixed-integer nonlinear program (MINLP) due to the complicating constraint (2).

The complexity of the model (P0) is more obvious if we look into the elaborate form of (2), as listed by (10)-(15). Two lower-level mixed-integer programs (MIPs) are nested in (P0) via (10)-(11), where the integer variables appear over the summation sign in a non-elementary form. The optimal solutions of the lower-level MIPs $\gamma_{b,t}$ and $\varphi_{b,t}$ appear below the summation sign in (12)-(13); moreover, (13) is a conditional constraint where the expression depends on the ordinal of the integer variables $\gamma_{b,t}$ and $\varphi_{b,t}$. Both (14) and (15) are nonlinear and nonconvex, while (15) is even a piecewise non-elementary function where the segments are pointed out by

the integer variables $\gamma_{b,t}$ and $\varphi_{b,t}$. Hence, the model (P0) is a BN-MINLP in the presence of (2), which is difficult to solve by resorting to canonical optimization theory.

Then, the situation is worse in the context of distributed dispatch, where only local information is available due to privacy concerns. Note that it is a mainstream practice of distributed dispatch to decompose a holistic dispatch problem into regional subproblems along with information exchange, as shown in [16, Sec. II-C]. However, due to the lack of global information, it is difficult for a distributed algorithm to find the optimal solution or even a feasible solution for the original holistic model if the model is too complicated. Most distributed algorithms rely on a convex and continuous model, while distributed optimization of MINLP remains a major technical challenge to date [25]. The existing distributed algorithms can not guarantee the convergence and feasibility of MINLP, let alone the much more complicated BN-MINLP, which contains a bi-level program. To the best of the authors' knowledge, distributed solution of BN-MINLP has never been reported.

III. TWO-STAGE ALTERNATING PROCEDURE FOR DISTRIBUTED DISPATCH

Since conventional optimization approaches do not apply to BN-MINLP, a two-stage alternating procedure is proposed here for decoupling the model (P0). The first stage tackles feasibility, while the second stage aims at optimality. The alternating procedure is embedded in each stage based on the following two observations:

- (1) The complexity of the model is significantly reduced if the mass flow $\mathbf{m}$ is temporarily fixed;
- (2) The DHN operator may optimize the mass flow $\mathbf{m}$ given a boundary condition $\mathbf{y}_E$ set by the electric power network operator.

A. Defending Feasibility in the First Stage

Feasibility is a critical issue in optimization, and it is difficult to find a feasible solution given the complexity of constraints in the BN-MINLP. Since the most complicated part, i.e., constraint (2), belongs to the DHNs, a relaxed BN-MINLP is investigated in this stage by introducing slack variables κ/v and $\omega \triangleq \text{col}(\omega_t)$ for thermal and hydraulic limits in DHNs:

$$(P1) \quad \zeta^* = \inf \quad \mathbf{I}^T \kappa + \mathbf{I}^T \mathbf{v} + \mathbf{I}^T \omega \quad (16)$$

$$\mathbf{x}_E, \mathbf{x}_H, \mathbf{y}_E, \mathbf{y}_H, \mathbf{m}, \xi, \\ \mathbf{Q}_H, \mathbf{d}_{HS}, \mathbf{F}_H, \mathbf{B}_H, \kappa, \mathbf{v}, \omega$$

$$\text{s.t.} \quad \underline{\tau} - \kappa \leq \tau \leq \bar{\tau} + \mathbf{v} \quad (17)$$

$$\mathbf{p}^s_{-1,t} - \mathbf{p}^r_{-1,t} \geq \underline{\delta}_t - \omega_t, \quad \forall t \quad (18) \\ \text{and (2)-(5), (7)-(8)}$$

which is still a BN-MINLP. If there is a feasible solution in (P0), then the optimum ζ^* in (P1) will be 0 (meaning no constraint violation) and the corresponding optimal solution will be feasible in (P0).

The main idea of the alternating procedure here is to solve (P1) by iteratively fixing part of the optimization variables. The procedure starts from the current state of the mass flow $\mathbf{m}^0$.

Given a fixed mass flow, the electric power network operator optimizes the boundary condition $\mathbf{y}_E$; given a fixed boundary state, the DHN operator optimizes the mass flow. Both energy sectors try to minimize the constraint violation in (16) at the first stage.

To be more specific, the electric power network operator solves the following linear program at the l -th iteration where ξ , $\mathbf{Q}_H$, $\mathbf{d}_{HS}$, $\mathbf{F}_H$ and $\mathbf{B}_H$ become constants inferred from $\mathbf{m}^{l-1}$:

$$(P1-E) \quad \begin{aligned} & \{x_E^l, x_H^l, y_E^l, y_H^l, \kappa_E^l, v_E^l, \omega_E^l\} \\ & = \arg \min_{x_E, x_H, y_E, y_H, \kappa, v, \omega} \mathbf{I}^T \boldsymbol{\kappa} + \mathbf{I}^T \mathbf{v} + \mathbf{I}^T \boldsymbol{\omega} \end{aligned} \quad (19)$$

$$\text{s.t.} \quad \mathbf{Q}_E \cdot \mathbf{y}_E + \mathbf{Q}_H^{l-1}(\mathbf{m}^{l-1}) \cdot \mathbf{y}_H \leq \mathbf{d}_{HS}^{l-1}(\mathbf{m}^{l-1}) \quad (20)$$

$$\mathbf{F}_H^{l-1}(\mathbf{m}^{l-1}, \xi^{l-1}) \cdot \boldsymbol{\tau} + \mathbf{B}_H^{l-1}(\mathbf{m}^{l-1}, \xi^{l-1}) \cdot h^{\text{EHC}} = \mathbf{d}_{TH} \quad (21)$$

$$\begin{cases} ps_{n1,t} - ps_{n2,t} = \mu_b(m_{b,t}^{l-1})^2 \\ pr_{n1,t} - pr_{n2,t} = \mu_b(m_{b,t}^{l-1})^2 \end{cases} \quad \forall t, \forall b: n1 \rightarrow n2 \quad (22)$$

and (3), (17)-(18)

where the optimum is denoted as ζ_E^l .

Subsequently, the DHN operator solves the following program to optimize the mass flow given the boundary condition $\mathbf{y}_E^l$:

$$(P1-H) \quad \begin{aligned} & \{m^l, \xi^l, x_H^l, y_H^l, \kappa_H^l, v_H^l, \omega_H^l\} \\ & = \arg \min_{m, \xi, x_H, y_H, \kappa, v, \omega} \mathbf{I}^T \boldsymbol{\kappa} + \mathbf{I}^T \mathbf{v} + \mathbf{I}^T \boldsymbol{\omega} \end{aligned} \quad (23)$$

$$\text{s.t.} \quad \mathbf{Q}_E \cdot \mathbf{y}_E^l + \mathbf{Q}_H(\mathbf{m}) \cdot \mathbf{y}_H \leq \mathbf{d}_{HS}(\mathbf{m}) \quad (24)$$

and (2), (5), (7)-(8), (17)-(18)

where the optimum is denoted as ζ_H^l .

Proposition 1 (On the Feasibility in the First Stage): The total violation of constraints is decreasing with the alternating iterations consisting of (P1-E) and (P1-H).

Proof: Firstly, since both (P1-E) and (P1-H) share the same set of constraints at the l -th iteration, the optimal solution in conjunction with the mass flow settings in (P1-E), i.e., $\boldsymbol{\beta} = \{m^{l-1}, \xi^{l-1}, x_H^l, y_H^l, \kappa_E^l, v_E^l, \omega_E^l\}$, will be also a feasible solution in (P1-H). Secondly, because ζ_H^l is the infimum of the objective function in (P1-H), we have $\zeta_H^l \leq \mathbf{I}^T \boldsymbol{\kappa}_E^l + \mathbf{I}^T \mathbf{v}_E^l + \mathbf{I}^T \boldsymbol{\omega}_E^l$ where the right-hand side is the objective value achieved by the feasible solution $\boldsymbol{\beta}$ in (P1-H). Thirdly, since both (P1-E) and (P1-H) share the same objective function, $\mathbf{I}^T \boldsymbol{\kappa}_E^l + \mathbf{I}^T \mathbf{v}_E^l + \mathbf{I}^T \boldsymbol{\omega}_E^l$ coincides with ζ_E^l , which leads to $\zeta_H^l \leq \mathbf{I}^T \boldsymbol{\kappa}_E^l + \mathbf{I}^T \mathbf{v}_E^l + \mathbf{I}^T \boldsymbol{\omega}_E^l = \zeta_E^l$. Following a similar technique, we can obtain $\zeta_E^{l+1} \leq \zeta_H^l$. Hence, $\zeta_E^{l+1} \leq \zeta_H^l \leq \zeta_E^l$, which means that the total violation of constraints is decreasing as the iterations proceed.

From Proposition 1, we know that the constraint violation is decreasing in the first stage. Suppose the constraint violation at the L -th iteration is small enough so that $\mathbf{I}^T \boldsymbol{\kappa}_E^L + \mathbf{I}^T \mathbf{v}_E^L + \mathbf{I}^T \boldsymbol{\omega}_E^L \approx 0$ with the mass flow setting $\mathbf{m}^{L-1}$ and heat dynamic setting ξ^{L-1} . ■

B. Approaching Optimality in the Second Stage

The feasible solution obtained in the first stage can be used as an initial point in the second stage, which aims for cost minimization as listed by the original model (P0).

Similarly, the alternating procedure is deployed here, which decomposes the model (P0) into two programs for the electric power network and the DHN respectively. The electric power network problem (P2-E) minimizes the total cost, while the DHN problem (P1-H) optimizes the DHN states to accommodate the boundary condition specified by the electric power network.

By fixing $\mathbf{m}$, the problem (P2-E) at the l -th iteration is listed as follows (where $l = L+1, L+2, \dots$):

$$(P2-E) \quad \begin{aligned} & \{x_E^l, x_H^l, y_E^l, y_H^l\} = \arg \min_{x_E, x_H, y_E, y_H} C(x_E, y_E, y_H) \\ & \text{s.t.} \quad (3), (6), (9), (20)-(22) \end{aligned} \quad (25)$$

where the optimum is denoted as C_*^l . This is also referred to as the CF-VT model adopted in large amounts of literature.

Proposition 2 (On the feasibility at the second stage): The model (P2-E) is feasible if the initial point $\{\mathbf{m}^L, \xi^L\}$ here is set as the final results in the first stage $\{\mathbf{m}^{L-1}, \xi^{L-1}\}$.

Proof: Now that $\mathbf{I}^T \boldsymbol{\kappa}_E^L + \mathbf{I}^T \mathbf{v}_E^L + \mathbf{I}^T \boldsymbol{\omega}_E^L \approx 0$ and $\{\mathbf{m}^L, \xi^L\} = \{\mathbf{m}^{L-1}, \xi^{L-1}\}$, $\boldsymbol{\beta}' = \{\mathbf{m}^L, \xi^L, x_H^L, y_H^L\}$ is obviously a feasible solution of the model (P2-E). Hence, there is at least a feasible solution for (P2-E), which means that (P2-E) is feasible.

The necessity of the first stage is highlighted via Proposition 2. Without the initial point yielded by the first stage, the CF-VT model (P2-E) can be hardly solved, not to mention the much more complicated VF-VT problem (P0). ■

Remark 1 (On the Mass Flow Rate and Necessity of Decomposition at the Second Stage): At this stage, the mass flow rate is still adjusted to approach the optimal solution in the second stage. Hence, decomposition is still necessary here.

C. Further Issues in Decomposition

We have shown the superiority of the two-stage alternating procedure via Propositions 1 and 2, which provide theoretical guidance. But the procedure still cannot be implemented in a distributed fashion yet. There are two critical issues left.

(1) At the electric power network side: the internal DHN variables x_H , the internal DHN constraints (6), (9), (17)-(18) and (20)-(22) and the internal DHN hydraulic/load settings $\mathbf{m}$, ξ , $\mathbf{Q}_H$, $\mathbf{d}_{HS}$, $\mathbf{F}_H$ and $\mathbf{B}_H$ are all collected in the electric power network subproblems (P1-E) and (P2-E). Privacy of the heat sector are all exposed to the electricity sector.

(2) At the DHN side: the non-elementary function is embedded via (2), making the DHN subproblem (P1-H) also a BN-MINLP, which is intractable according to the analysis in Section II-B.

Distributed implementation will be discussed, and tractable electric power network and DHN subproblems will be formulated in the following two sections respectively.

IV. ELECTRIC POWER NETWORK SUBPROBLEM FORMULATION

To preserve information privacy and reduce the scale of the dispatch problem, a sequential network equivalent for the DHN under VF-VT is proposed, which projects the DHN feasible region defined by both thermal and hydraulic constraints onto the boundary. Then a reduced electric power network subproblem is formulated, where the electric power network operator updates the boundary condition y_E according to the heat equivalent model.

A. Sequential Network Equivalent for District Heating Network With Variable Mass Flow

It is very difficult to directly perform network equivalent on the DHN under VF-VT, which is a BN-MINLP. Instead, noting that the mass flow is fixed iteratively in the electric power network subproblems (P1-E) and (P2-E), we develop a sequential network equivalent method here.

First, the equivalent technique is conducted given a fixed mass flow m^{l-1} . According to the results in [12], the thermal equations (5) detailed in (57) and (66)-(73) of Appendix A can be readily reduced:

$$\tau = \left[F_H^{l-1} \left(m^{l-1}, \xi^{l-1} \right) \right]^{-1} \times \left[-B_H^{l-1} \left(m^{l-1}, \xi^{l-1} \right) \cdot h^{EHC} + d_{TH} \right] \quad (26)$$

where the nonsingularity of the matrix $F_H^{l-1}(m^{l-1}, \xi^{l-1})$ is proved by [12, Sec. III]. Interested readers may refer to Appendix A in [12] for the elaborate form of (26). The equivalent technique maps the internal states of the DHN to the boundary states, which protects the private information of the DHN. As proved in [12], it accurately reflects the dynamics of DHN by using the heat network equivalent technique. Since it is based on Gaussian elimination, it does not involve solving other subproblems.

However, VF-VT has introduced the hydraulic equation (22), which is not handled by existing literature and thus warrants specific investigation here. To protect the privacy of the DHN in the solution procedure, this method will eliminate the DHN internal variables ps_{-1} / pr_{-1} in the hydraulic equations. The key step is that we choose arbitrarily from the pipelines in the DHN a spanning tree that stems from the heat station, and then there is a unique path that directs the mass flow from the heat station to each heat load in the supply pipes. A small example in Fig. 10 in Appendix B in the supplementary material clarifies the multiple choices of selecting such a spanning tree. Please note that most DHNs are radial [26], where we do not even bother to choose a spanning tree.

By using the incidence matrix, (22) can be reformulated into a matrix form:

$$\begin{cases} ps_{-1,t} = I \cdot ps_{1,t} - T_{-1,t} \cdot \text{diag}(\mu_b) \left(m_t^{l-1} \circ m_t^{l-1} \right) \\ pr_{-1,t} = I \cdot pr_{1,t} + T_{-1,t} \cdot \text{diag}(\mu_b) \left(m_t^{l-1} \circ m_t^{l-1} \right) \end{cases}, \quad \forall t. \quad (27)$$

$T_{-1,t}$ is the corresponding path-branch incidence matrix where the heat station is treated as the reference node and thus

eliminated [27]. I represents an all-one vector with an appropriate dimension inferred by the context. μ is the pressure loss factor. $\circ$ denotes the Hadamard product and $\text{diag}(\cdot)$ generates a diagonal matrix from the specified elements. In mathematics, the Hadamard product is the componentwise product of matrices. It produces a new matrix, where each element i, j is the product of elements i, j of the original two matrices. The reformulation is based on two important characteristics: 1) In a meshed network, the pressure loss equation for a link is actually redundant; 2) The node pressure in a DHN are dependent on each other, so the pressure of a reference node can be selected as an independent variable. Appendix B provides more information on how the reformulation is achieved physically.

The mass flow m will be updated by the DHN subproblem, and then the next round of network equivalent begins by fixing the mass flow as m^l .

Remark 2 (On the Equivalent DHN Model): The equivalent DHN model is not a simplification but an accurate equivalent reduction, and thus it does not compromise the exactness of the solution. We use the equivalent DHN model mainly to protect the privacy of DHN, which consists of the thermal part and the hydraulic part. Essentially, Gaussian elimination is adopted during the equivalent procedure of the two parts. Hence, the exactness is guaranteed.

B. Reduced Electric Power Network Subproblem

With the equivalent technique proposed above, the DHN thermal/hydraulic feasible regions in electric power network subproblems (P1-E) and (P2-E) are projected onto the boundary variables by plugging (26)/(27) into (17)/(6) and (18)/(9) respectively. Then the electric power network subproblems (P1-E) and (P2-E) in both stages can be reduced as the following subproblems (P1-Er) and (P2-Er):

$$\begin{aligned} \text{(P1-Er)} \quad & \{x_E^l, y_E^l, y_H^l, \kappa_E^l, v_E^l, \omega_E^l\} \\ & = \arg \min_{x_E, y_E, y_H, \kappa, v, \omega} I^T \kappa + I^T v + I^T \omega \end{aligned} \quad (28)$$

$$\begin{aligned} \bar{\tau} - \kappa & \leq \left[F_H^{l-1} \left(m^{l-1}, \xi^{l-1} \right) \right]^{-1} \\ & \times \left[-B_H^{l-1} \left(m^{l-1}, \xi^{l-1} \right) \cdot h^{EHC} + d_{TH} \right] \leq \bar{\tau} + v \end{aligned} \quad (29)$$

$$\begin{aligned} I \cdot (ps_{1,t} - pr_{1,t}) & \geq \underline{\delta}_t - \omega_t + 2T_{-1,t} \\ & \cdot \text{diag}(\mu_b) \left(m_t^{l-1} \circ m_t^{l-1} \right), \quad \forall t \end{aligned} \quad (30)$$

and (3), (20)

$$\text{(P2-Er)} \quad \{x_E^l, y_E^l, y_H^l\} = \arg \min_{x_E, y_E, y_H} C(x_E, y_E, y_H) \quad (31)$$

$$\begin{aligned} \bar{\tau} & \leq \left[F_H^{l-1} \left(m^{l-1}, \xi^{l-1} \right) \right]^{-1} \\ & \times \left[-B_H^{l-1} \left(m^{l-1}, \xi^{l-1} \right) \cdot h^{EHC} + d_{TH} \right] \leq \bar{\tau} \end{aligned} \quad (32)$$

$$\begin{aligned} I \cdot (ps_{1,t} - pr_{1,t}) & \geq \underline{\delta}_t + 2T_{-1,t} \\ & \cdot \text{diag}(\mu_b) \left(m_t^{l-1} \circ m_t^{l-1} \right), \quad \forall t \end{aligned} \quad (33)$$

and (3), (20)

where the DHN internal variables $\mathbf{x}_H$ and parameters have been all eliminated or encoded. The reduction here is exact while the privacy of the DHN is well preserved. For instance, the EPN operator cannot infer the DHN topology (i.e., the sparsity structure of $\mathbf{T}_{-1,t}$) from the numerical results of $\delta_t + 2\mathbf{T}_{-1,t} \cdot \text{diag}(\mu_b)(\mathbf{m}_t^{l-1} \circ \mathbf{m}_t^{l-1})$.

Remark 3 (On the Internal DHN Units): Due to the complexity of VF-VT, in this paper, we only consider electricity-heat coupling units including CHP units, heat pumps and water pumps at the boundary between the electric power network and DHN. For the cases with internal DHN units such as fuel-based heat generation instead of CHP, (P2-Er) can be further decomposed by using the modified Benders decomposition algorithm proposed in our previous research [13]. Then the total cost can be minimized in a distributed fashion, where the DHN operator does not disclose any information of the internal units to the EPN operator.

V. DHN SUBPROBLEM FORMULATION

The DHN subproblem (P1-H) is intractable in the presence of the non-elementary function (2) introduced by the node method. To devise a practical solution, we investigate a surrogate model to describe the heat dynamics. Then (P1-H) is approximated as a continuous nonlinear program by replacing the node method equations with the surrogate thermal dynamic equations, which can be readily solved by commercial solvers.

A. Surrogate Thermal Dynamic Model

Basically, the node method is developed by discretizing the solution of the following partial differential equation [28] that describes the thermal dynamics:

$$c\rho A_b \frac{\partial \tau}{\partial t} + cm \frac{\partial \tau}{\partial x} + \lambda_b (\tau - \tau^A) = 0, \quad \forall b \quad (34)$$

where x measures the distance from the starting end of pipeline b and τ^A denotes the ambient temperature. Note that (34) is not discretized yet and therefore there are no subscripts b and t in the mass flow temperature variable τ and the mass flow variable m .

To remove the non-elementary function (2) and the complicating dynamic vector ξ , a surrogate thermal dynamic model is derived here by approximating the partial derivatives in (34) with a first-order difference:

$$\begin{cases} c\rho A_b \left[0.5(\tau_{b,t}^{PS,in} + \tau_{b,t}^{PS,out}) - 0.5(\tau_{b,t-1}^{PS,in} + \tau_{b,t-1}^{PS,out}) \right] \\ + cm(\tau_{b,t}^{PS,out} - \tau_{b,t}^{PS,in})/L_b + \lambda_b [0.5(\tau_{b,t}^{PS,in} + \tau_{b,t}^{PS,out}) - \tau^A] = 0, \forall b, t \\ c\rho A_b \left[0.5(\tau_{b,t}^{PR,in} + \tau_{b,t}^{PR,out}) - 0.5(\tau_{b,t-1}^{PR,in} + \tau_{b,t-1}^{PR,out}) \right] \\ + cm(\tau_{b,t}^{PR,out} - \tau_{b,t}^{PR,in})/L_b + \lambda_b [0.5(\tau_{b,t}^{PR,in} + \tau_{b,t}^{PR,out}) - \tau^A] = 0, \forall b, t \end{cases} \quad (35)$$

where we approximate $\partial \tau / \partial t$ by referring to the average temperature of pipeline b , and $\tau_{b,t}^{PS,in} / \tau_{b,t}^{PS,out}$ and $\tau_{b,t}^{PR,in} / \tau_{b,t}^{PR,out}$ denote mass flow temperature at the inlet/outlet of pipeline b at period t , respectively. The approximation error can be alleviated by introducing fictitious nodes for long pipelines, which mathematically reduces the differential step length.

Then we can replace the node-method model (specified by equations (2) here and (70)-(73) in Appendix A) with the surrogate model in (35). In this way, the complicating dynamic vector ξ and the non-elementary function (2) are removed, and the constraint (5) is approximated by the following constraint:

$$\mathbf{F}'_H(\mathbf{m}) \cdot \boldsymbol{\tau} + \mathbf{B}'_H(\mathbf{m}) \cdot h^{\text{EHC}} = \mathbf{d}'_{\text{TH}} \quad (36)$$

The approximation in (35) does have errors, but its merits have been proved in [20], and we also validate that its accuracy is also acceptable in our tests. In addition, a more reliable and generic approximation of the dynamics of DHNs (essentially a set of partial differential equations in (34) is a critical yet unresolved challenge in DHN modeling research. This topic is so fundamental even in centralized analysis, and thus is apparently out of the scope of this single paper that focuses on the distributed solution. Note that alternative approximation may be utilized apart from (35). The surrogate model here can use other models, as long as it is easy to solve relative to BN-MINLP. In the two-stage alternating procedure, we only borrow this model to optimize the mass flow, convergence, and feasibility judgment are still based on the accurate node method.

B. Approximate DHN Subproblem

With the help of the surrogate model, an approximate DHN subproblem can be readily listed:

$$\begin{aligned} \text{(P1-Ha)} \quad & \{m^l, \mathbf{x}_H^l, \mathbf{y}_H^l, \boldsymbol{\kappa}_H^l, \mathbf{v}_H^l, \boldsymbol{\omega}_H^l\} \\ & = \arg \min_{\mathbf{m}, \mathbf{x}_H, \mathbf{y}_H, \boldsymbol{\kappa}, \mathbf{v}, \boldsymbol{\omega}} \mathbf{I}^T \boldsymbol{\kappa} + \mathbf{I}^T \mathbf{v} + \mathbf{I}^T \boldsymbol{\omega} \quad (37) \\ \text{s.t.} \quad & (24), (36), (7)-(8) \text{ and } (17)-(18) \end{aligned}$$

which is a continuous nonlinear program that can be solved by existing commercial solvers such as IPOPT.

C. Whole Iterative Procedure

From the discussion above, the whole two-stage alternating procedure embedded with sequential equivalent techniques is summarized in Fig. 2, where the information exchanged between the electric power network operator and the DHN operator is highlighted in blue.

The reduction conducted in the electric power network subproblems (P1-Er) and (P2-Er) is exact, since the techniques used here are problem reformulation (with the help of spanning trees), variable substitution and Gaussian elimination. Both (P1-Er) and (P2-Er) essentially solve (P0) with constant flow, whereas the private information of DHNs is concealed in the projected feasible region.

Although a surrogate thermal dynamic model is borrowed to formulate the DHN subproblem (P1-Ha), this does not compromise the feasibility of the proposed procedure. Both the feasibility and the convergence are finally checked and validated by the electric power network subproblems (P1-Er) and (P2-Er), respectively, which are both exact. Also, the effectiveness will be shown numerically in the next section.

Remark 4 (on Propositions 1 and 2): Note that (P1-E), (P1-H) and (P2-E) in Propositions 1 and 2 are not used in the solution procedure, but as mentioned in Section IV-A, (P1-Er)

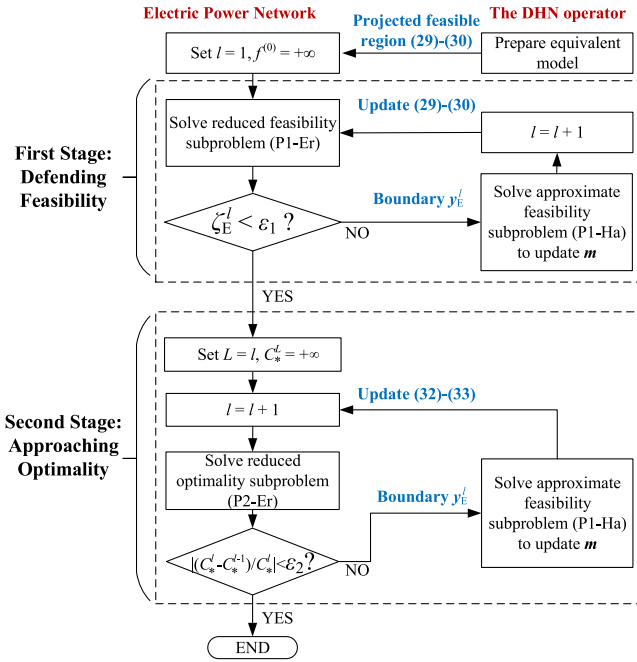


Fig. 2. Flowchart of the proposed two-stage alternating procedure.

and (P2-Er) are equivalent to (P1-E) and (P2-E), respectively. Hence, Propositions 1 and 2 also prove that a distributed solution comprising of (P1-Er), (P1-H) and (P2-Er) has strict feasibility in an ideal case where an oracle can figure out the optimal solution of (P1-H). Propositions 1 and 2 play the role of theoretical guidance and show the advantages of the two-stage alternating procedure. In addition, as mentioned in the third paragraph of Section III-C, (P1-H) is still a BN-MINLP, which cannot be directly solved by the existing optimizer. Hence, we use an approximate version (P1-Ha) which can be solved by IPOPT, as a workaround for engineering practice. We believe this does not compromise the theoretical value of Propositions 1 and 2.

VI. SIMULATION RESULTS

In this section, case studies are conducted on two IEHSs of different scales, which are derived from [18]. Details of system parameters are all available online [29]. Since integrated dispatch under CF-VT has been extensively shown to be superior to the separated heat-driven dispatch [5], [7], [9], [10], [11], [12], [13], this section focuses on the most flexible operation of DHNs, namely, VF-VT. Particularly, we demonstrate the benefits brought by VF-VT, the accompanying computational challenges and how the proposed distributed solution manages to deal with these issues. To this end, we compare the proposed distributed solution (abbreviated as “distributed VF-VT” hereafter) with the following two methods:

1) CF-VT method [5], [7], [9], [10], [11], [12], [13] (abbreviated as “CF-VT” hereafter): The model adopted here can be readily yielded from (P0) by treating the mass flow $\mathbf{m}$ as parameters, which is equivalent to (P2-E) according to the discussion in Section II-B. Gurobi is adopted to solve the model.

TABLE I
ECONOMIC PERFORMANCE IN THE SMALL-SCALE SYSTEM

Dispatch method	CF-VT	VF-VT
Wind curtailment penalty (\$)	2,854	579
Electric power network dispatch cost (\$)	99,908	97,470
DHN dispatch cost (\$)	10,723	10,418
Total cost (\$)	113,485	108,467

2) Centralized solution to the VF-VT model [18] (abbreviated as “centralized VF-VT” hereafter): Based on the fix-point method, the key iterations in [18] consist of two steps. In the l -th iteration, the first step is to set ξ^l according to the mass flow $\mathbf{m}^{l-1}$ and the non-elementary function in (2). Then the second step is to solve the ξ -fixed model, which is derived by removing constraint (2) and setting ξ as ξ^l in (P0). The algorithm converges when the increment $\Delta\xi = \|\xi^l - \xi^{l-1}\|_\infty$ is small enough, e.g., less than 0.1. Obviously, the constraint (2) and the remaining constraints in (P0) have been separately considered in the first and the second steps. Therefore, this method cannot guarantee feasibility, as we will show later.

Case studies are organized as follows. First, distributed VF-VT is compared with CF-VT in [5], [7], [9], [10], [11], [12], [13], which defends the necessity and motivation of this research. Then, the drawbacks of the centralized VF-VT [18] are investigated, while we show the superiority of the proposed distributed VF-VT in feasibility guarantee and privacy protection. Next, the convergence of distributed VF-VT is analyzed. Then we test distributed VF-VT in a system with more coupling units to prove the generality of this method. Finally, a practical-scale provincial IEHS in [18] is tested to further highlight the scalability of distributed VF-VT. All tests are run on a computer with an Intel i7-8750 CPU and 8 GB of RAM. The electric power network subproblems (P1-Er) and (P2-Er) are solved by the Gurobi solver, while both the DHN subproblems (P1-Ha) in distributed VF-VT and the ξ -fixed model in centralized VF-VT are solved by the interior point method-based IPOPT solver. In distributed VF-VT, the convergence tolerances ε_1 and ε_2 are set as 10^{-5} and 1% respectively by default. We are following the initialization method in [18], which includes two methods. One is a warm start using historical data and the other is a flat start. Since we do not have historical data, we use the flat start method directly.

A. Small-Scale System

This system is composed of a six-bus electric power network and a six-node DHN [18]. There are 2 non-CHP units, a CHP unit, a water pump, 3 heat loads and a wind farm. Unless specified, the results of VF-VT are obtained via the proposed distributed VF-VT method by default. The mass flow of the heat station is always 250 kg/s in CF-VT, while it varies within the range of 240-400kg/s in VF-VT. The upper limit of the temperature of all nodes in the DHN is 120 °C and the lower limit is 60 °C.

1) *Economic Benefits of VF-VT*: VF-VT is first compared with CF-VT to analyze its economic benefits, which is summarized in Table I. As shown in Fig. 3 (a), VF-VT reduces

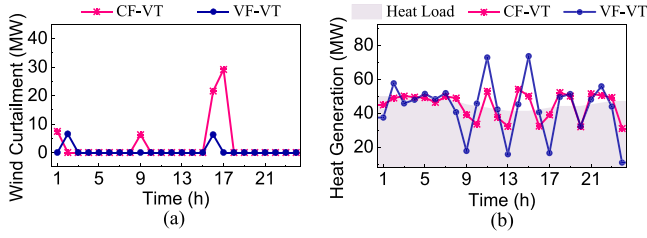


Fig. 3. Comparison of optimization results between CF-VT and VF-VT: (a) wind curtailment, and (b) heat generation.

penalty cost for wind spillage by 79.7%, which contributes to a final 4.4% reduction in the total cost.

To be more specific, the flexibility of mass flow adjustment in VF-VT helps relax the implicit technical constraints on the output of CHP units. Take period 17 as an example to facilitate analysis where wind power suddenly grows to 106.6 MW and the improvement of VF-VT is the most prominent. In CF-VT, the return side temperature of node 4 with the heaviest heat load is 65.79°C at period 16, which is close to the lower bound (60°C). Consequently, the heat output of CHP has to be increased to keep the temperatures above the lower limit at period 17, as shown by the red line with asterisks in Fig. 3 (b). Due to the electricity-heat coupling, the electricity output of CHP in period 17 also rises to 8.41 MW by 28.0% compared with the previous period, which squeezes the room for wind power utilization and results in a wind curtailment of 29.14 MW. On the contrary, VF-VT makes it possible to reduce the output of the CHP unit at period 17 due to the following two reasons. On one hand, at period 15, both the heat generation and the mass flow are magnified (where the mass flow in the heat station reaches 393.4 kg/s) so that the DHN has stored more energy in advance. On the other hand, in period 17, VF-VT reduces the mass flow in the heat station to 240.0 kg/s, which relieves the impact of heat generation reduction on the temperature drop at the supply side. Both the adjustments of mass flow before and at period 17 help enlarge the allowable variation range of the output of the CHP unit, contributing to more accommodation of wind power and lower total costs.

2) *Robustness of VF-VT*: Then the robustness of CF-VT and VF-VT is compared via a sensitivity analysis in the presence of heat load variation. To quantify the extent of infeasibility, an infeasible gap is defined as the minimal total violation of all constraints. Intuitively, a zero infeasible gap infers that the solution is strictly feasible, while a larger infeasible gap indicates more critical threats will be posed to the IEHS with the given dispatch solution. In CF-VT, this can be obtained by the optimum of the relaxed version of (P2-E), i.e., (P1-E). In VF-VT, this can be calculated by the optimum ζ^* of the relaxed version of (P0), i.e., (P1), where the control variables are fixed to the dispatch decisions yielded by different solution methods such as distributed VF-VT here.

The infeasible gap of CF-VT grows exponentially with increasing heat load, while VF-VT strictly maintains system security, as shown in Fig. 4 (a). Take the case of a 20% heat load growth rate as an example. In CF-VT, the temperatures at nodes 3, 4 and 6 at the return side significantly violate the

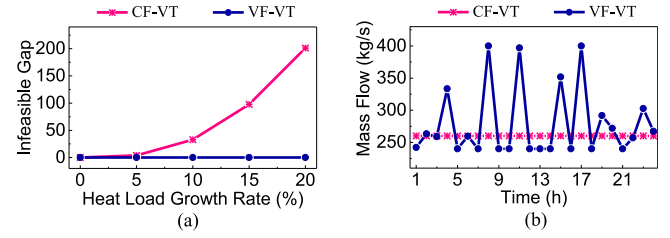


Fig. 4. Comparison between CF-VT and VF-VT in the presence of heat load variation: (a) infeasible gap, and (b) mass flow with a 20% heat load growth rate.

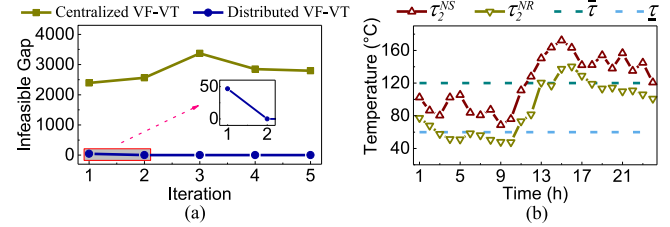


Fig. 5. (a) Infeasible gap of centralized VF-VT and distributed VF-VT, (b) temperature of node 2 in centralized VF-VT.

lower limit (60°C). At node 4 (which is a heat load node), for instance, this kind of violation is observed at 19 out of 24 periods, where the temperature at the return side reaches 49.7 °C at period 1 with a 10.3 °C violation of the lower limit. In contrast, VF-VT manages to deal with this issue with the help of variable mass flow, as shown in Fig. 4 (b). During period 1, VF-VT reduces the mass flow in the heat station by 7.9 kg/s (250.0 kg/s in CF-VT, 242.1 kg/s in VF-VT) and thus increases the temperature at the supply side given a certain heat generation. Besides, VF-VT increases the mass flow at node 4 by 21.0 kg/s (110.0 kg/s in CF-VT, 131.0 kg/s in VF-VT) and thus alleviates the temperature drop at this heating load node. Based on the above two adjustments, VF-VT manages to pull the low temperatures in CF-VT back within the safe operating limits.

3) *Advantages of Distributed VF-VT Solution*: Although the VF-VT model (P0) is favorable in cost minimization and robust operation from the above discussion, it can be hardly solved by the existing method, i.e., centralized VF-VT in [18]. A full DHN model has to be collected by the electric power network operator to implement centralized VF-VT, which infringes the privacy of the heat sector. Since the information exchanged between different energy sectors only involves boundary information (e.g., boundary states y_E^l and boundary constraints (29)/(32) and (30)/(33)), this privacy issue is well addressed by distributed VF-VT.

A more critical issue of centralized VF-VT relates to feasibility. As shown by the green lines with squares in Fig. 5 (a), the infeasible gap of centralized VF-VT remains in the order of thousands, indicating a significant constraint violation, which is not relieved as the iterations proceed. The reason is that centralized VF-VT fails to jointly consider the complicating constraint (2) and the remaining constraints in (P0), which threatens the safe operation of the IEHS. As shown in Fig. 5 (b), a few tens of degrees Celsius of upper-limit (120°C) violation is observed at a single period (e.g., for 13

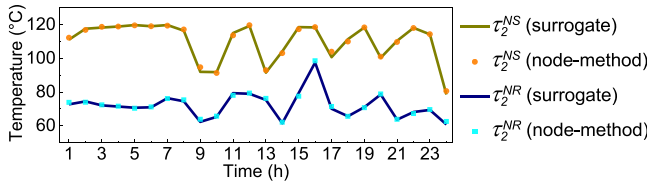


Fig. 6. Comparison of the temperature of Node 2 between the proposed surrogate model and node-method model.

$\leq t \leq 23$) at a single node (i.e., node 2). The pipes of DHN will be easily damaged if the operating temperature constantly falls out of the operational limit, which not only shortens their service life, but also probably leads to engineering disasters. In contrast, the infeasible gap of distributed VF-VT, as shown by the blue lines with circles in Fig. 5 (a), falls drastically to 0, which validates Propositions 1 and 2 numerically. Note that the x-axis label “iteration” in Fig. 5 (a) and subsequent figures refer to total iterations index l . The first stage of distributed VF-VT consists of the first and second iterations, and the second stage begins with the third iteration. It requires only two iterations for the first stage to find a feasible solution. And then the infeasible gap remains 0 in the whole second stage until convergence at the 5th iteration, which means that all constraints hold and the feasibility of the solution is strictly guaranteed. Hence, distributed VF-VT manages to guard the security of the IEHS under the flexible yet complicated VF-VT operation.

Besides, better economic performance is observed in distributed VF-VT. Note that there is no optimum in the cases where the dispatch solutions are infeasible. For a fair comparison, the total costs of infeasible cases are evaluated by adding the operation costs $C(\mathbf{x}_E, \mathbf{y}_E, \mathbf{y}_H)$ and the penalty costs $\eta\zeta^*$ for constraint violation, where the penalty price η is set as \$10 in the simulation. The total costs of distributed VF-VT and centralized VF-VT are \$108,467 and \$133,269, respectively. As for the computational performance, 5 iterations cost distributed VF-VT 47.86 seconds (versus 43.86 seconds elapsed in centralized VF-VT).

4) *Accuracy Validation of Surrogate Thermal Dynamic Model:* The rationale for adopting the surrogate model in distributed VF-VT is to be verified here. The simulation is conducted by investigating whether the surrogate model in Section V-A can capture the true thermal dynamic of the exact node-method model that prevails in existing literature (e.g., refs. [7], [8], [10], [12], [13], [24]). The results obtained by both the surrogate and the exact models are visualized in Fig. 6 (taking Node 2 as an example), where the temperatures from both models are very close. On the whole, the average temperature error of all nodes at all times is 0.925 °C, which is an acceptable level. Note that these errors do not necessarily compromise the feasibility of distributed VF-VT. The surrogate model is used to facilitate the solution of the DHN subproblem, while the feasibility is checked by the electric power network subproblem where the DHN feasible region is projected based on the exact node-method model.

5) *Convergence Analysis:* The influence of tolerances ε_1 and ε_2 on the convergence and accuracy of distributed VF-VT is

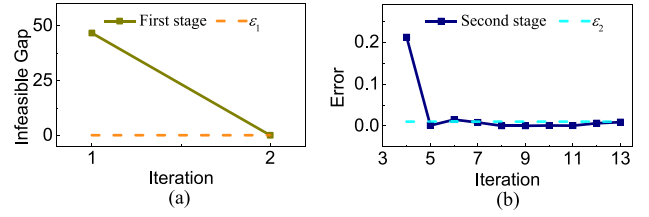


Fig. 7. Convergence curve of distributed VF-VT: (a) first stage, and (b) second stage.

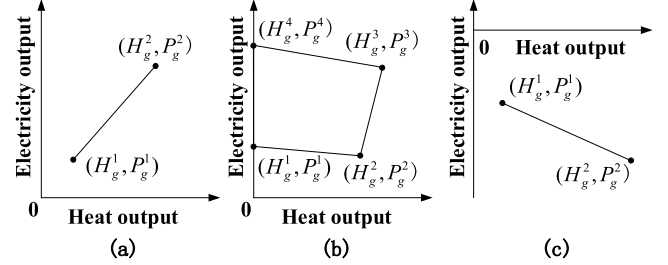


Fig. 8. Operating region of (a) back-pressure CHP, (b) extraction-condensing CHP and (c) heat pump.

studied by the analysis of the iterative convergence process. To get a full convergence curve, we did not terminate the iteration in Fig. 7 even when the error hits the tolerance ε_2 at the 5th iteration.

Fig. 7 shows that only two iterations are required to obtain a feasible solution in the first stage. Since the infeasible gap drops drastically to 0, and the tolerance ε_1 in the first stage does not significantly affect the convergence. As shown in Fig. 2, we take the relative error in objective between two adjacent iterations where C_*^L is initialized as $+\infty$, and therefore the error of the 3rd iteration (i.e., the first iteration at the second stage) is 0. But this does not infer convergence at the 3rd iteration, and thus the error at this iteration is not depicted in Fig. 7 (b) to avoid misunderstanding. The error remains at a 10^{-3} level after convergence at the 5th iteration, which is acceptable.

6) *Higher Coupling Level System:* To test distributed VF-VT in a higher coupling level, we further add a DHN heated by a large-scale heat pump and 3 distributed heat pumps. Details are all available in [29]. Distributed heat pumps generally have small power, while large-scale heat pumps usually have a larger heat production capacity.

The operating region of an electricity-heat coupling unit is described by a polyhedron in (51)-(52), which is general and flexible. For example, as shown in Fig. 8, the switch between CHP and heat pump can be achieved by simply changing the number and locations of the extreme points of the polyhedron.

Unlike a large-scale heat pump which is connected to the “root” of the primary pipeline network for district heating, the small-scale distributed heat pump is usually connected to the “leaves” such as household heat consumers, which are widely distributed in the secondary pipeline network [30], [31], and [32]. In reality, heat exchange stations serve as an intermediary to transfer heat from primary pipelines to secondary ones, which means that the total heat

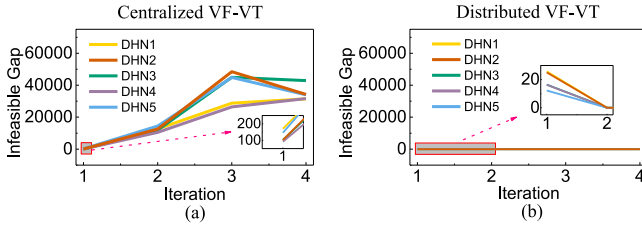


Fig. 9. Comparison of the evolution of the infeasible gap in the practical scale system between (a) centralized VF-VT, and (b) distributed VF-VT.

demand of secondary pipelines is summed as an aggregated load described by (57) in the heat exchange station model. Besides, there are few practical measuring points in secondary pipelines [10]. Hence, modeling of secondary pipelines is usually not considered in CF-VT research [8], [10], [12], [13]; note that the constant flow assumption is relaxed here, which has already rendered the VF-VT problem much more challenging due to the nonconvexity and discreteness. Considering the problem complexity and the huge difference of the heat power between primary pipelines and distributed heat pumps, the impact of distributed heat pumps on the primary pipelines is temporarily neglected, and the heater model in our previous work [33] is adopted to describe distributed heat pumps:

$$T_{h,t}^{in} = T_{h,t-1}^{in} + \alpha_h(T_{h,t}^{out} - T_{h,t}^{in}) + \beta_h d_{h,t} \quad (38)$$

$$\underline{T}_h^{in} \leq T_{h,t}^{in} \leq \bar{T}_h^{in} \quad (39)$$

where $d_{h,t}$ is the electric power consumption of distributed heat pump in house h at period t . $T_{h,t}^{in}$ and $T_{h,t}^{out}$ represent the indoor and outdoor temperature of house h at period t , respectively. α_h and β_h are the thermal coefficients of the environment and the distributed heat pump of house h . The coupling between the thermal dynamics of house h and the electric power consumption of distributed heat pump is described by (38). Constraint (39) specifies the temperature range which is comfortable for end-users.

Simulation shows that, in this case, it takes distributed VF-VT 4 iterations in total within 44.22 seconds to converge. Compared with CF-VT, the total operating cost of the system is reduced from \$231,377 to \$114,120, which further shows that VF-VT has better economic benefits.

B. Practical-Scale System

The scalability of distributed VF-VT is further verified in a practical-scale provincial IEHS where the electric power network is interconnected with 5 DHNs [18]. There are 319 buses, 431 branches, 60 non-CHP units, 5 CHP units and 34 wind farms in the electric power network, while there are 40 nodes, 35 pipelines, 5 water pumps and 20 heat loads in total in the DHNs.

As shown in Fig. 9, the infeasible gap enlarges to a few tens of thousands in centralized VF-VT, while the infeasible gap in distributed VF-VT diminishes to 0 within 2 iterations. At the end of the 4th iteration, distributed VF-VT converges with a total cost of \$318,912 (versus \$2,051,326 in

TABLE II
ECONOMIC PERFORMANCE UNDER DIFFERENT LOADING CONDITIONS IN THE PRACTICAL-SCALE SYSTEM

Heat load growth rate	2%	4%	6%	8%	10%
Total cost (\$)	320,060	321,298	323,215	326,790	331,815

its centralized counterpart). Since the solution of the centralized VF-VT in the practical-scale system is also infeasible, the total infeasible gap of 174,279 (p.u.) for the 5 DHNs resulted in a penalty cost of \$1,742,790 (penalty price η is set as 10 \$/p.u.), which accounted for 84.96% of the total cost (\$2,051,326). The total 4 iterations cost distributed VF-VT 62.47 seconds (versus 6,149.77 seconds elapsed in its centralized counterpart). The benefits brought by decomposition are twofold. First, the 5 DHN subproblems (P1-Ha) can be solved in parallel. Second, the problem scale is much smaller in distributed VF-VT. There are 2,760 variables and 2,856 constraints in total in the largest (P1-Ha) in distributed VF-VT, while there are 20,856 variables and 33,504 constraints in the ξ -fixed model in centralized VF-VT. Hence, the scalability of distributed VF-VT is significantly superior.

Then the robustness of distributed VF-VT under the variation of heat load in the practical-scale system is studied by sensitivity analysis. Table II shows that as the heat load grows, the total cost of the system increases accordingly. When the heat load increases to 10%, distributed VF-VT can still find a feasible solution to ensure the safety of the system. Distributed VF-VT improves the flexibility and robustness of the system by adjusting the mass flow.

VII. CONCLUSION

Due to the distributed physical structure and privacy concerns in IEHSs, distributed dispatch is more practical to achieve coordinated IEHS operation. However, to facilitate decomposition, the mass flow has been generally assumed to be constant in distributed IEHS dispatch research, resulting in a CF-VT operation. Relaxing such assumptions leads to an IEHS dispatch problem under VF-VT, which enhances flexibility and optimality while posing unresolved technical challenges. The resultant problem is essentially a BN-MINLP, which is not handled by any distributed solution to the best of the authors' knowledge. Hence a two-stage alternating procedure is proposed here for distributed IEHS dispatch under VF-VT, namely, distributed VF-VT.

Numerical results first consolidate the necessity of investigating IEHS dispatch under VF-VT. The advantages of VF-VT over CF-VT are twofold. On one hand, VF-VT manages to lower the total costs of the IEHS in the base case operation. On the other hand, in the presence of demand variation, VF-VT is more robust by adjusting the mass flow, while system security cannot be maintained by CF-VT.

Then, comparative studies are extended to reveal the issues of the existing solution method to the resultant BN-MINLP,

i.e., centralized VF-VT. Distributed VF-VT clearly distinguishes itself from the centralized VF-VT in the following ways:

(1) Privacy preservation: Global information has to be collected to reach the final dispatch decision in centralized VF-VT, while boundary information is exchanged to promote coordination among energy sectors in distributed VF-VT.

(2) Feasibility guarantee: The infeasible gap resulting from centralized VF-VT remains large, indicating significant operation limit violation. On the contrary, distributed VF-VT can strictly ensure a safe operation of IEHSs, which is theoretically proved by Propositions 1 and 2 and numerically verified in case studies.

(3) Scalability: The computation burden substantially increases with the scale of IEHSs in centralized VF-VT, which may even lead to divergence. In contrast, distributed VF-VT remains highly efficient in a practical-scale IEHS by dividing the original BN-MINLP into smaller electric power network/DHN subproblems. The average computation time for each iteration therein is 15.62 seconds in distributed VF-VT (versus 1537.44 seconds in centralized VF-VT). The efficiency of distributed VF-VT is compatible with intraday dispatch, where the dispatch interval normally varies from 15 minutes to 1 hour.

How to reach the global optimum of the BN-MINLP is an interesting yet open challenge, which goes beyond the discussion of a single paper and thus is left as future work. In addition, in the future, we will also study how to build a distributed model from scratch, consider the internal heat source of DHN, develop an accurate model to approximate the thermal dynamics under VF-VT, and propose a better warm start method.

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Qingju Luo, photograph and biography not available at the time of publication.